

VOIDSCOPE · REQUIREMENTS

VoidScope requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Round-2 decision 3A (VDS-DDR-004): water offered with the tip steered aside; medical lead's confirmation in drills. Requirement wording unchanged
0.4	2026-10-03	Amish Chadha	R11 restated by Amish (decision 3A) as seeing around a 45-degree bend from a straight hole, with a test-box verification; met on paper by the steerable camera tip (VDS-DDR-003); status of R1, R2, R4, R8 and R9 updated from VDS-CAL-001 v0.2
0.3	2026-10-03	Amish Chadha	TRL 3: status from VDS-CAL-001 on the constructable design (VDS-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2: R2 split into reach (R2) and bends (R11); R5 restated as a ceiling with a bite valve; R6 adds the blocked-tip limit; R8 worded as a value-engineering target; R12 two-way voice added; concept status for each
0.1	2026-09-30	Amish Chadha	Initial scaffold

VoidScope requirements

Eleven of the twelve requirements are met on paper by the constructable design with its steerable camera tip: eight by calculation and three by design, all to be confirmed by test at TRL 4. R9 (setup in 3 minutes) is estimated at 2.2 minutes but can only be settled by timed drills. R11 was restated on 2026-10-03 by Amish, who chose option A for it ("3A", in his reply "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"): instead of following a channel with two 45° bends, the probe goes in straight and its camera tip, steered from the handle, looks around a 45° bend. The rigid rod, the air path through its bore and the water tube to the bite valve are unchanged.

Table 1. Requirements and status at TRL 3 (tags in brackets refer to VDS-CAL-001).

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Probe passes a standard core hole	Outer diameter of tip and rod at or below 50 mm	Pass the assembled probe through a 51 mm test ring along its full length	Met with the tip locked straight: 48 mm at the head and tip, 47 mm at the grip [A1]. The handle-end fittings (the tail cap's air barb and now the steering control) are wider and stay outside the hole
R2	Reach into a void along a straight path	Working length of at least 4 m (13 ft)	Measure assembled length from the grip to the tip	Met: 4,203 mm from the grip; 4,131 mm can go in before the steering control reaches the hole [A4g, A4]
R3	See in darkness and dust	Read a 20 mm high letter at 1 m in a dark, dusty test box	Test box with a set dust load and no ambient light	Met on paper: 19 pixels per letter against 10 needed; 7 lx left after 2 m of dusty air [C2, C6]; dust load to test
R4	Survive rubble handling	Head survives 20 drops of 1 m onto concrete and 1 m water immersion for 30 minutes	Drop and immersion tests followed by a video check	By design: IP68 camera in an aluminium tip housing, sheathed bending section, nose with 2.85 mm minimum wall [G2]; to test, with the tip straight and bent
R5	Controlled water delivery	Drip from 0 up to a ceiling of no more than 5 mL/min at any allowed height up to 40 °C; no flow when the clamp is closed or the valve is not bitten	Timed collection at the tip at set heights and temperatures	Met: ceiling 1.5 mL/min at 1 m and 20 °C, 4.5 mL/min at 2 m and 40 °C [D1, D2]
R6	Air delivery to the void	At least 20 L/min of filtered air at the tip; never more than 1 kPa above ambient, even with the tip blocked	Flow meter and manometer at the tip, open and blocked	Met: 44 L/min at full speed, 0.9 kPa at shut-off [E3, E5]
R7	Shift-long battery life	At least 4 h of continuous video and light at 20 °C and at -10 °C	Run-down test at 20 °C and at -10 °C	Met: 6.2 h and 4.3 h on one pack [F4]
R8	Cost against the value-engineering target	Full kit parts cost against the USD 2,000 target	Costed bill of materials from named suppliers	USD 1,780, USD 220 under the target [I2, I4]
R9	Quick to deploy	Two trained people assemble and start searching within 3 minutes	Timed drills with a USAR partner	Estimate 2.2 min [H6]; only a timed drill can settle it

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R10	Hygienic water path	Water path parts replaceable in under 5 minutes without tools	Timed replacement against a cleaning procedure	By design: single-use water set, about 4 min [D8]; to time
R11	See around a 45-degree bend from a straight hole (restated 2026-10-03, Amish, decision 3A)	With the probe pushed straight through a 51 mm hole, steer the camera to look down a passage that turns 45° to either side and read what is there	Test box: a 51 mm hole into a straight passage 300 mm wide that turns 45° 0.5 m beyond the hole. Push the probe straight in, steer the tip from the handle and read 20 mm letters 1 m down the turned passage on the monitor; repeat to the other side; straighten the tip and withdraw it through the hole	Met on paper: the tip bends up to 90° each way [J3]; bent 45° it needs 83 mm to the side [J6] and a 20 mm letter at 1 m gets 19 pixels [C2]
R12	Talk with the survivor	Two-way voice between the operator and a person at the tip	Speech test in a rubble box	By design: tip speaker as speaker and microphone, push-to-talk headset

■ Assumptions

- Local crews can open or find gaps of about 50 mm into most voids of interest.
- A rigid, sectioned rod is good enough for a first version, with a camera tip that bends in one plane; turning the rod chooses the plane. Following a bent channel with the rod itself is not required (R11 as restated).
- A slow gravity drip through a bite valve is an acceptable way to give water when a medical lead directs it; the bite valve stays on the nose and the operator steers the tip aside to give water (drill card, build plan section 5a); the medical lead's confirmation in drills is a TRL 4 item.
- Commodity sealed pipe-inspection camera heads meet the image, light and sealing targets.

Safety: These requirements describe rescue equipment that delivers water and air to a trapped person and carries a lithium battery. The steerable tip must be locked straight before it goes into or comes out of a hole. Meeting them on paper does not make the design safe to use. VoidScope is an open engineering reference, not certified rescue equipment and not a medical device; see VDS-PRC-001, Safety.